File System and Storage Management

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Date Submitted: September 10, 2026

Ubuntu Server

A. Partition & Mount

Device Name: /dev/sdb1

File System Type: ext4

Mount Point: /projectdata

Commands used:

```
sudo fdisk /dev/sdb       # create partition (n, defaults, w)
sudo mkfs.ext4 /dev/sdb1
sudo mkdir /projectdata
sudo mount /dev/sdb1 /projectdata
```

Command Outputs:

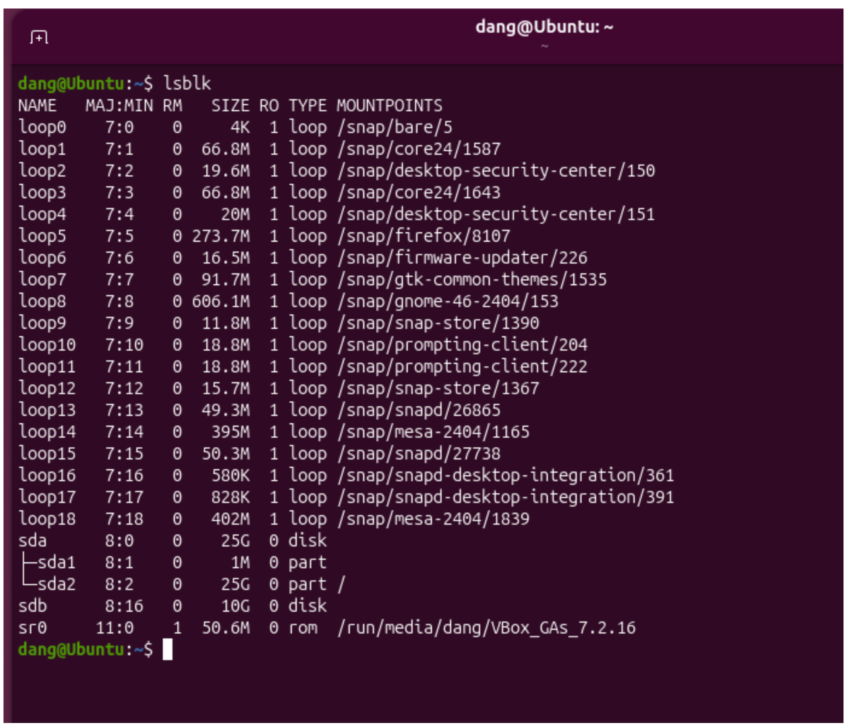
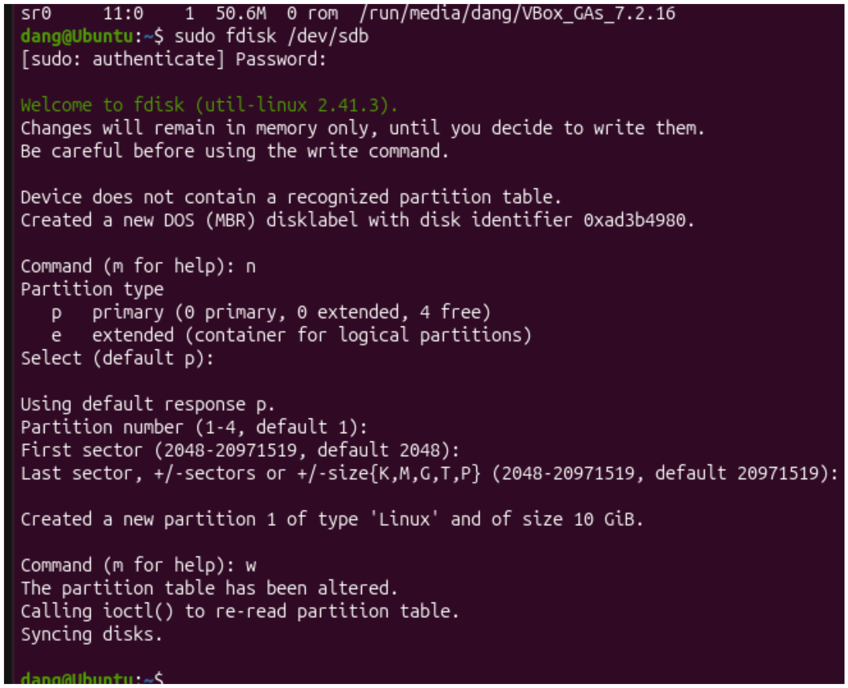
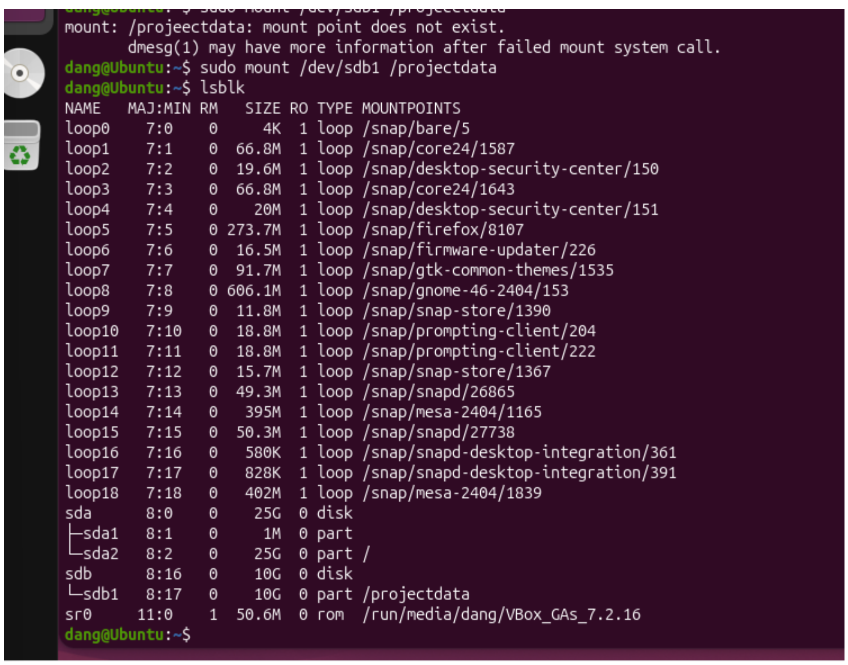


Figure 1. lsblk confirms the new 10 GB disk (/dev/sdb) is detected, unpartitioned.



```
sr0     11:0    1  50.6M  0 rom  /run/media/dang/VBox_GAs_7.2.16
dang@Ubuntu:~$ sudo fdisk /dev/sdb
[sudo: authenticate] Password:

Welcome to fdisk (util-linux 2.41.3).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0xad3b4980.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-20971519, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20971519, default 20971519):

Created a new partition 1 of type 'Linux' and of size 10 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

dang@Ubuntu:~$
```

Figure 2. Creating partition /dev/sdb1 using fdisk.



```
mount: /projeectdata: mount point does not exist.
       dmesg(1) may have more information after failed mount system call.
dang@Ubuntu:~$ sudo mount /dev/sdb1 /projectdata
dang@Ubuntu:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0     7:0    0     4K  1 loop /snap/bare/5
loop1     7:1    0  66.8M  1 loop /snap/core24/1587
loop2     7:2    0  19.6M  1 loop /snap/desktop-security-center/150
loop3     7:3    0  66.8M  1 loop /snap/core24/1643
loop4     7:4    0    20M  1 loop /snap/desktop-security-center/151
loop5     7:5    0 273.7M  1 loop /snap/firefox/8107
loop6     7:6    0  16.5M  1 loop /snap/firmware-updater/226
loop7     7:7    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop8     7:8    0 606.1M  1 loop /snap/gnome-46-2404/153
loop9     7:9    0  11.8M  1 loop /snap/snap-store/1390
loop10    7:10   0  18.8M  1 loop /snap/prompting-client/204
loop11    7:11   0  18.8M  1 loop /snap/prompting-client/222
loop12    7:12   0  15.7M  1 loop /snap/snap-store/1367
loop13    7:13   0  49.3M  1 loop /snap/snapd/26865
loop14    7:14   0   395M  1 loop /snap/mesa-2404/1165
loop15    7:15   0  50.3M  1 loop /snap/snapd/27738
loop16    7:16   0   580K  1 loop /snap/snapd-desktop-integration/361
loop17    7:17   0   828K  1 loop /snap/snapd-desktop-integration/391
loop18    7:18   0   402M  1 loop /snap/mesa-2404/1839
sda       8:0    0    25G  0 disk
├─sda1    8:1    0     1M  0 part
└─sda2    8:2    0    25G  0 part /
sdb       8:16   0    10G  0 disk
└─sdb1    8:17   0    10G  0 part /projectdata
sr0      11:0    1  50.6M  0 rom  /run/media/dang/VBox_GAs_7.2.16
dang@Ubuntu:~$
```

Figure 3. Formatting, mounting, and lsblk confirming /dev/sdb1 mounted at /projectdata.

B. Directory Setup & Permissions

Directory Path: /projectdata

Group Assigned: facultygrp (owner, read/write), studentgrp (ACL, read-only)

Permission Level: 770 on facultygrp via chmod, plus an ACL granting studentgrp read+execute (r-x); others have no access

Commands used:

```
sudo chown root:facultygrp /projectdata

sudo apt install acl -y

sudo setfacl -m u::rwx,g::rwx,g:studentgrp:rx,o::--- /projectdata
```

Command Outputs:

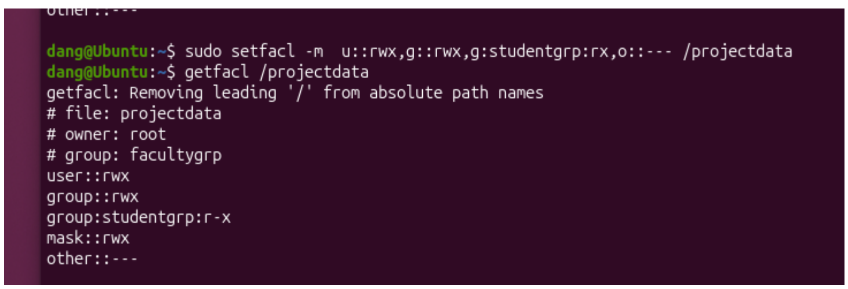


Figure 4. getfacl output confirming facultygrp has rwx and studentgrp has r-x, with others denied.

C. Verification

Tested User	Action	Access Allowed	Access Denied
faculty2	touch (write)	Yes ✓	—
student4	ls (read)	Yes ✓	—
student4	touch (write)	—	Yes ✓

Command Outputs:

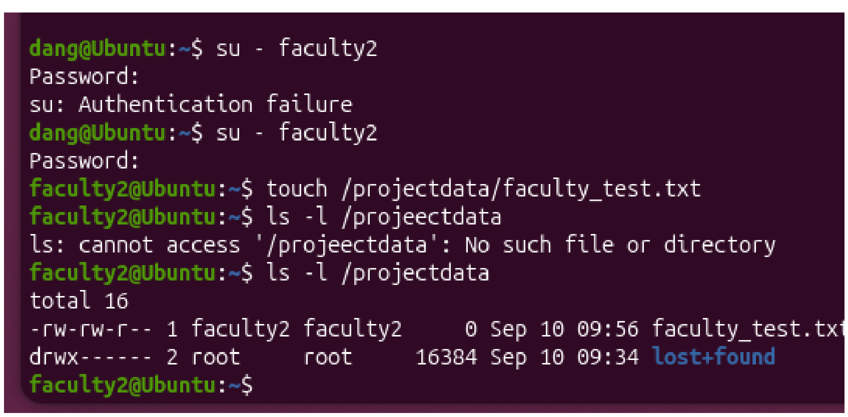


Figure 5. faculty2 successfully creates a file in /projectdata.

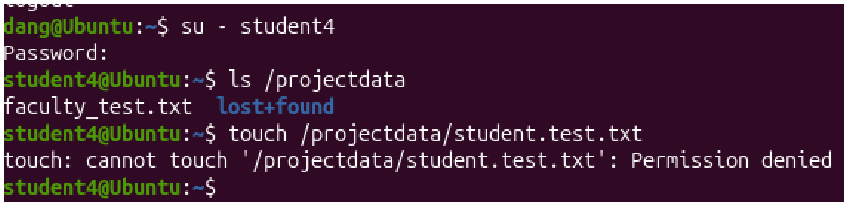


Figure 6. student4 can list /projectdata (read) but is denied when attempting to write.

Part 3: Reflection

What challenges did you encounter in creating partitions and managing permissions?

Partitioning and mounting went pretty smoothly, but the permissions part took me a few tries. I accidentally typed chown 770 instead of chmod 770 at one point, which set the owner to a weird value instead of actually changing the permission bits. Once I caught that and fixed it, chmod still wasn't giving me the result I expected, turns out there was already an ACL sitting on the folder from an earlier command, and that was capping the group permissions even after I changed them. What finally worked was setting all the ACL entries in one single command instead of piecing them together, then double-checking with getfacl to make sure facultygrp and studentgrp each had the right level of access on the same folder.

How can these skills be applied in real-world system administration duties?

This is basically what a sysadmin would deal with when setting up shared storage for a company or school, adding a new disk when a department needs space, then making sure people only get the access they actually need. ACLs come in handy especially when a single folder needs to serve more than one group differently, like faculty and students sharing the same space but needing different levels of access. That kind of setup comes up a lot in schools or offices with several teams working off shared drives.